



From Aggregates to Estimates - population inference from MPD and standard data sources

Module 2 - Session 2.4.4

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MPD data: non-probability samples

**MNO subscribers are
not a random sample of the population,
nor can be assumed to be.**

Big Data: only as “big” as their population coverage

Harvard Professor of Statistics Xiao-Li Meng has published on [biases of Big Data](#):

“Which one should I trust more: a 1% survey with 60% response rate or a self-reported administrative dataset covering 80% of the population?” (...)

“The “bigness” of such Big Data (for population inferences) should be measured by the relative size $f = n/N$, not the absolute size n ; when combining data sources for population inferences, those relatively tiny but higher quality ones should be given far more weights than suggested by their sizes.” (...)

“This should remind us that, without taking data quality into account, population inferences with Big Data are subject to a Big Data Paradox: the more the data, the surer we fool ourselves.” (...)

“Compensating for quality with quantity is a doomed game.” (...)

“It is therefore essentially wishful thinking to rely on the “bigness” of Big Data to protect us from its questionable quality, especially for large populations.”

(Meng 2018: 685ff)

Biases are different for each statistic

- Bias/error is a measure of a statistic/parameter, not of a data source
- In other words: there is no biased data source, but there are biased parameters/statistics

Definition of bias

There are two types of errors

- Random error
- Systematic error or bias

In general terms, the bias of an estimate is the difference between the estimate and the true value. For a mean estimate of a variable x , the bias is simply the difference between the estimate and the true mean:

$$\text{bias}(\hat{\mu}_x) = \hat{\mu}_x - \mu_x.$$

Whenever general population parameters are estimated directly from samples of mobile phone users (excluding non-users in general or also users of MNOs not covered), the bias of such estimates is the difference between these estimates and the true population parameters. For the mean, this can be expressed as:

$$\text{bias}(\hat{\mu}_{x, \text{phoneusers}}) = \hat{\mu}_{x, \text{phoneusers}} - \mu_x.$$

“Measurement” and representation biases

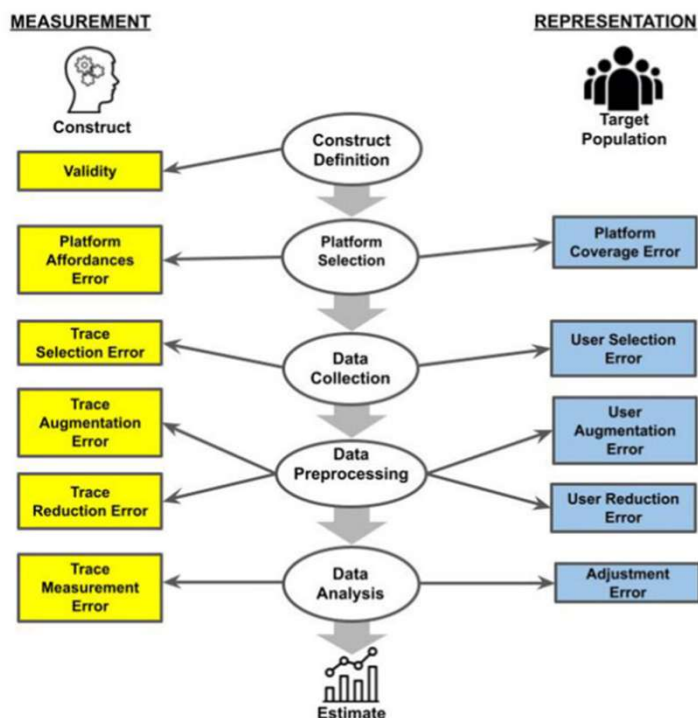


Figure 2. Potential measurement and representation errors in a digital trace-based study lifecycle. Errors are classified according to their sources:

(Sen et al. 2021)

Estimates may include biases on the “measurement” side, related to

- location & mobility operationalisation
- data errors & artefacts
- subsetting (e.g. discarding cells)
- data gaps
- ...

As well as biases regarding representation, stemming from

- population coverage of mobile phones
- particular MNO coverage (if not all MNOs)
- subsetting (e.g. user activity)
- ...

Every data processing step may also introduce biases.

Big Data alone are not the solution - and of limited use

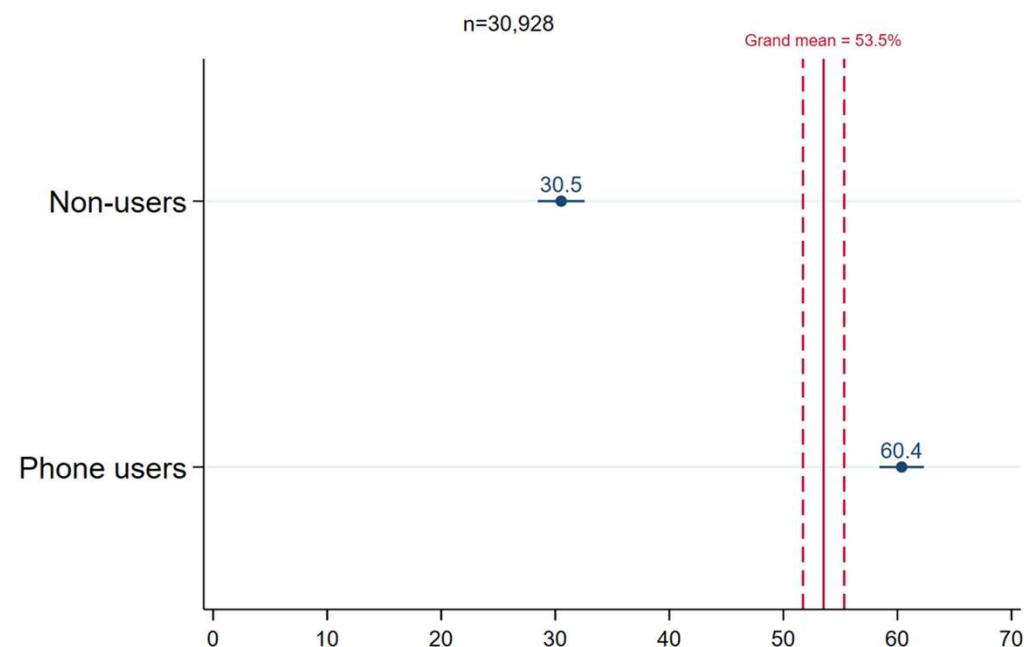
- Statistics derived from MPD are often inherently biased, and need to be bias-adjusted for population inference
- Big Data (MPD) are only as “big” as their population coverage
- Bias-assessment and adjustment require standard data sources; Big Data cannot replace standard data sources (censuses, surveys, registers), but only complement them
- Every context (country, region, district) is different, there is no standard model, no standard parameter, no automation
- Bias-adjustment cannot reduce biases to zero, but only mitigate it
- Unbiased small-area time-series estimates are statistically utopian, given current data sources and methods

Biases due to population coverage

Mobility is often different between mobile phone users and **non-users**

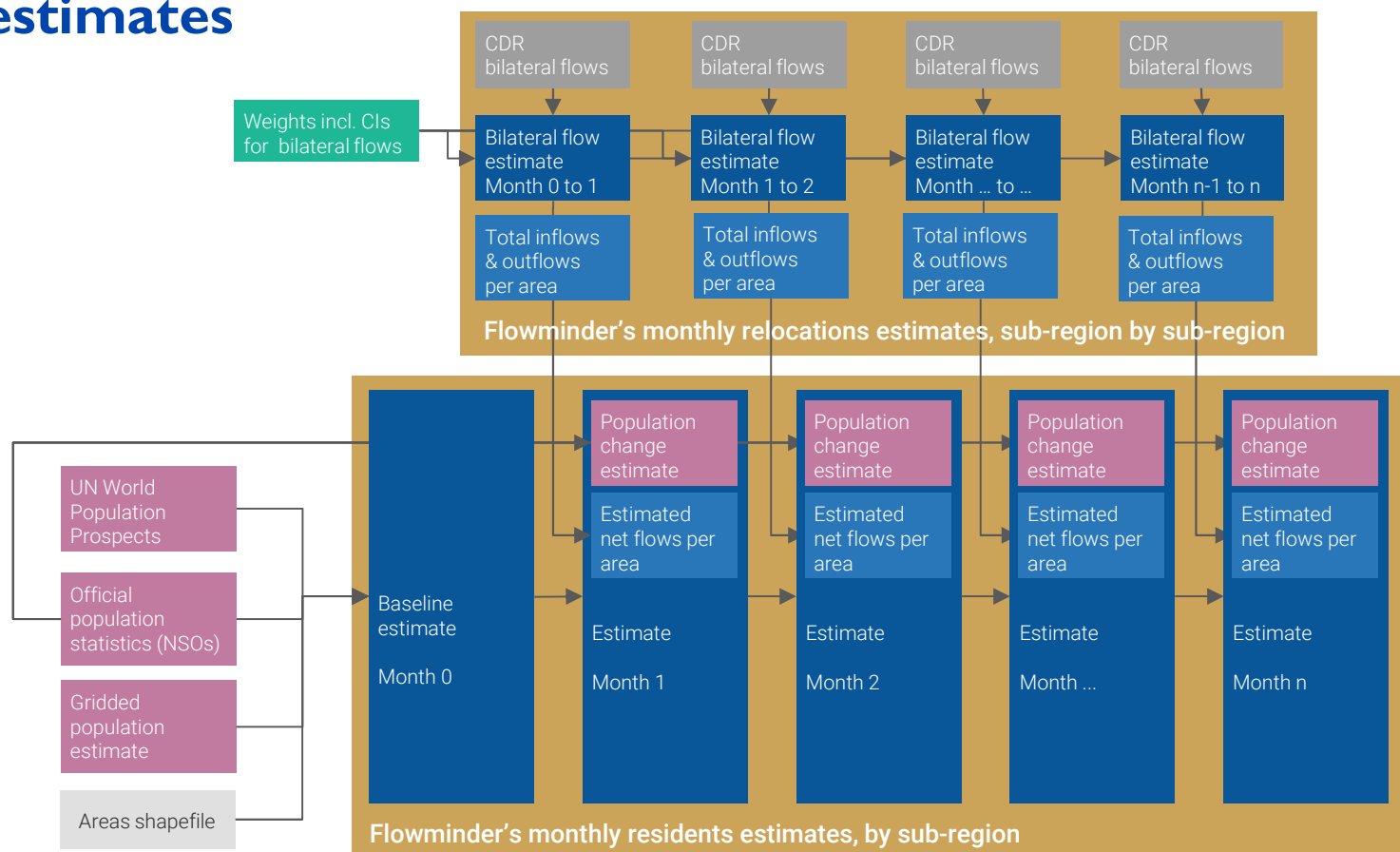
- With mobile phone users showing **higher mobility**, on average, **than non-users**
- Mobility estimates based on phone user data alone often **overestimate** (at times: underestimate) **mobility**
- Not all mobility indicators, however, show such large differences between these two groups

Ghana: % of population who did a 5+km trip in past 3 months



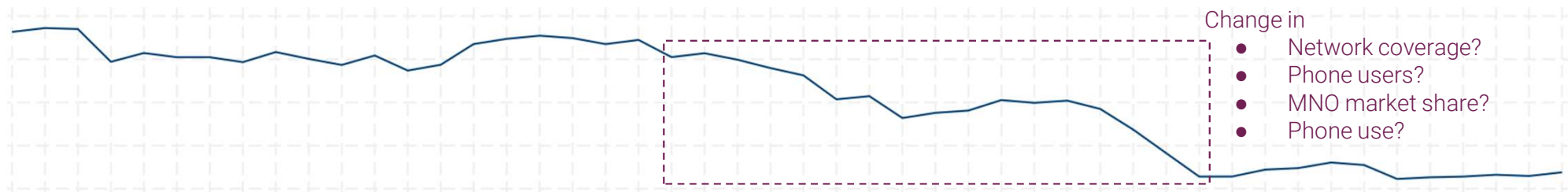
Source: AHIES 2022, Q3, weighted (pop_weight)

Use case population statistics: a method for monthly flow and stock estimates

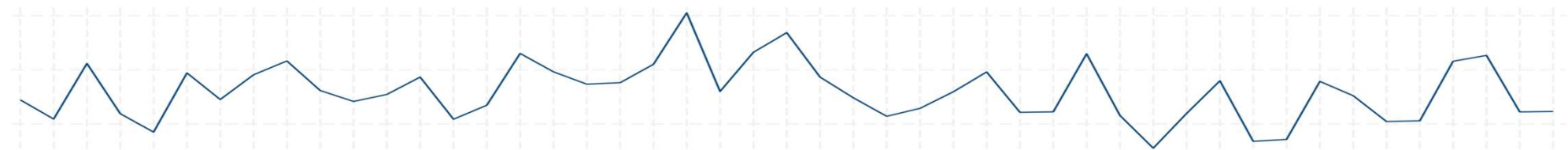


Use case population statistics: weighting stocks or flows?

- Aggregates of (locatable) IDs per month (**stocks**) show substantial fluctuations over time, not necessarily indicative of population change



- Aggregates of **flows** of (locatable) IDs per month are more stable, less affected by changes in MNO shares, cell outages, etc. This is why we use **flows for estimating changes in stocks**, rather than scaling stocks.



Bias-adjustment: domain-level re-weighting (a naive estimator)

- As suggested by (BERĘSEWICZ et al. 2018), a linear correction of the **domain-level parameter** $\check{\theta}_d$ measured in Big Data, accounting for
 - The population coverage (a_d^{cover}) per domain
 - The active users (a_d^{active}) per domain
 - The share of the particular data source among all providers (a_d^{share}) per domain

$$\check{\theta}_d^{adj} = \check{\theta}_d \times a_d^{cover} \times a_d^{active} \times a_d^{share}$$

- Drawbacks of domain-level weighting:
 - No correction for biases in θ_d due to selection mechanism!
 - Coverage per domain may be unknown and need to be estimated (-> **surveys required**)
 - Market share per domain may be unknown and need to be estimated (-> **surveys required**)

Domain-level re-weighting of CDR flow data

- Flow estimates are adjusted for
 - The count of MSISDNs rather than users in CDR relocation aggregates
 - The selection error of counting MSISDNs of MNO users, not the entire population

$$\text{est_flow}_{ab} = \text{cdr_flow}_{ab} * \text{geomean}(\text{mno_sims}_a, \text{mno_sims}_b)^{-1} * \text{median_6m}(\text{geomean}(\text{pop_coverage}_a, \text{pop_coverage}_b)^{-1})^{\text{att}}$$

- The parameters used are
 - The number of SIMs per user (mno_sims_x) per area x (x=a for origin, x=b for destination)
 - The population coverage of the CDR data (pop_coverage_x) per area x
 - An attenuation factor (att) for weights (power shrinkage)

Population statistics: general caveats



Relative change estimates strongly depend on **baseline population number** - differs greatly between data sources for LMICs



Statistical limits to **granularity of survey-based weights** - separate weights for admin3-to-admin3 flows require very large survey sample sizes or even census data



Longitudinal survey data are needed



Lack of **validation data**: mobility and population estimates at admin3 are rare for LMICs



In areas with **very low coverage**, flows might be greatly underestimated

General overview of estimation approaches

When the aim is to use Call Detail Records for inferences about a larger population - say of a country, a region, a district; of all persons irrespective of age and gender - the CDR data need to be combined with other available data sources. This is usually referred to as **data triangulation, data fusion, data integration** or similar terms.

The related statistical problems are also discussed under the umbrella term *Big Data*, and are similar to problems that have been discussed in the methodological survey literature and statistical literature in recent decades. The problem of data integration of Big Data and standard data sources is **akin to the problem of combining probability samples and non-probability samples** (Kim 2022).

So, which estimation approaches are available for population inferences based on CDR data and other data sources such as surveys, censuses or other auxiliary data (Rao 2020)?

General overview of estimation approaches

When there is a non-probability sample (such as from CDR data) and a probability sample (such as from survey data), three main scenarios can be distinguished for data integration (Rao 2020):

I. Target variable observed in both samples

Ideally, if both the non-probability sample and the probability sample include measures of the target variable (y), estimates from both samples can be combined via data integration estimators (Kim and Tam 2018), also discussed as hybrid estimates (Tam et al. 2022)

I. Target variable not observed in the probability sample (but common auxiliary variables)

If there is a probability sample (A) available that includes auxiliary variables also included in the non-probability sample (B), propensity scores can be calculated from sample A and a ratio estimator can be used to estimate y from B (Kim and Wang 2019). Doubly robust estimators have also been suggested (Kim and Wang 2019).

I. Only non-probability sample available (but known population totals for auxiliary variables)

If it is possible to find suitable statistical predictors of the target variable y by using some auxiliary variables x , this prediction model can be applied to the entire population.

General overview of estimation approaches

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COMBINING ESTIMATES (Y)

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ESTIMATING Y VIA X

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I. Only non-probability sample available (but known population totals for auxiliary variables)

FINDING PREDICTORS (X)

If it is possible to find suitable statistical predictors of the target variable y by using some auxiliary variables x , this prediction model can be applied to the entire population.

Which data do you have?

If you have been using or are using Big Data for a statistical estimate, do you have ...

- I. **Target variable (Y) observed in both samples**

- I. **Target variable not observed in the probability sample (but common auxiliary variables X)**

- I. **Only non-probability sample available for Y (but known population totals for auxiliary variables X)**

- I. **Other setup**

General overview of estimation approaches

The authors of the report on methodologies in EUROSTAT's MNO-MINDS project distinguish three main families of approaches (Zhang et al. 2024):

I. Randomisation approach

Design-based approach, no statistical model; requires dedicated survey data, parameters taken from survey data

Current use cases are tourism statistics for Indonesia. The added value of MNO data under this approach is efficiency and reduced sample size requirements

Of limited use, because the survey parameters are target-specific, and will vary across space, time and topic, and "bias cannot be avoided practically in the randomisation approach to MNO data".

The estimates also depend on the availability of the survey-based parameters, which are usually low granularity.

Example given:

$$N = w * m$$

N ... number of tourists

m ... number of active-roaming foreign SIMs (CDR aggregate)

w ... weight calculated from a dedicated survey, including several parameters (deduplication of devices, proportions of tourists among travellers, proportion of roamers among tourists)

Note: This is the same approach discussed as **domain-level re-weighting** by Beresewicz et al. 2018

General overview of estimation approaches

The authors of the report on methodologies in EUROSTAT's MNO-MINDS project distinguish three main families of approaches (Zhang et al. 2024):

II. Quasi-randomisation

Model-based approach; models the underlying selection mechanism *as if* the MNO data were observed based on known probabilities; post-stratification is one example;

Could remove the need for specialised surveys, if a suitable model can be identified

Firstly, it requires **access to the CDR microdata and to survey or census microdata** with mobile phone use or ownership as a variable

If the selection mechanism can be modelled from these data, estimation then only requires CDR aggregates as well as population totals from the Statistical Office

General overview of estimation approaches

The authors of the report on methodologies in EUROSTAT's MNO-MINDS project distinguish three main families of approaches (Zhang et al. 2024):

III. Super-population modelling (statistical modelling)

Here, different models are needed for each target statistic.

- A. Data fusion: based on the **assumption** that the target distribution is the same as that of the input data (big data); the error cannot be quantified, **external validation** is required to make the outputs usable as official statistics
- B. MNO features for **prediction**: target variable is available from non-MNO data, such as survey or census data; feasibility of producing statistics depends on the predictive power and size of the errors
- C. Geographically weighted regression: based on MNO counts as **proxy variables**; non-parametric methods

Planning data initiatives - scoping of data and methods

- Every statistic, every estimate is different ...
 - For feasibility studies, pilot projects, data exploration: in addition to assessing CDR aggregates, there's a need to assess all other available data sources in parallel, before committing to specific estimates
 - For use cases: assess the statistical feasibility of estimates, given the available data sources, before committing to provide any statistic
- ... and the bias-adjustment approach will be different for each statistic

Selected literature

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The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).





Mobile Phone Data for Policy Programme (WB-GDF)
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